

Computer Vision 1

Course Manual 2024

Martin Oswald, Arnoud Visser
Qi Bi

University of Amsterdam

08-2024

1 Overview

This is a 6 credit course in 7 weeks, with a final exam in week 8. The study material is given by the lecture slides, background papers and relevant state of the art literature (all readings provided on Canvas).

The course offers the following services:

- Theoretical **lectures** based on the freely downloadable book <http://szeliski.org/Book/> ('Szeliski'). 2022 edition.
- Weekly **seminars** are given to practice the basic theory and assist you with solving exercises.
- **Practical lab sessions** will be held that will help you with completing the assignments.

The lectures, seminars and lab sessions will all be held on-site on Science Park.

2 Lectures

The lectures will be mostly held by Arnoud and Martin except for some guest lectures in the last week before the exam (TBD). The lectures will cover the exam material, and will provide a theoretical understanding of basic Computer Vision techniques, that you will apply during the lab assignments.

While the lectures will be recorded and the recordings shared afterwards, we highly recommend attending the lectures in person. We cannot be held responsible for faulty recordings or recording equipment.

3 Seminars

Every week (except the first), seminars will be held on Tuesdays where theoretical exercises will be discussed. These exercises are a good preparation for the exam. The exercises will be published in advance, so you can prepare these well before the seminars and we encourage you to solve them as much as possible on your own. Solutions to the exercises will be published after the seminars.

3.1 Seminar preparation: lecture quiz

As a preparation for the exercise seminars we require you to complete a lecture quiz. A total of 5 quizzes will be conducted via Canvas, from week 2 to 6. The purpose of these quizzes is to test your knowledge of the lecture material. The quizzes are comprised of a few (4-6) questions. They are pass/fail: you get a passing grade for each quiz if you answer a specified number of questions correctly. This number differs between quizzes.

Passing these quizzes determines a small part of your final grade (see Grading section). Completing the quizzes is not mandatory, but highly recommended.

4 Practical lab sessions

There will be four lab assignments and a final project.

All assignments will be done in groups of two or three students. These groups will be pre-assigned by the course administrators (see 'People' tab on Canvas). Every group member should contribute equally to completing each assignment.

Each assignment consists of programming tasks and open-ended questions, and is designed to provide you hands-on experience with modelling computer vision architectures.

The submission dates for the assignments will be provided in each assignment instruction, and can be found in the Course Overview on Canvas.

The weekly on-site lab sessions are organised to work on the lab assignments and to ask the TA's for support.

In addition, in the first week of the course an introduction to Python will be provided. This introduction will not be graded and has no contribution to your final grade. This lab assignment is useful for brushing up your Python skills and concerns some basic image processing and Numpy techniques.

Secondly we will offer a self-check in the required mathematics (see Canvas). Be sure to take a look at this checklist to gauge your background knowledge for successfully completing the course.

5 Lab assignment quality demands

For the first four lab assignments you will complete and submit an iPython notebook. We will assess and grade your running code, as well as your understanding of the

theoretical framework that accompanies the code.

For the final project you will submit your code and a written report in a zip-file.

For full credit, make sure your report follows these guidelines:

- Submit your notebooks with the following naming convention: **studentID1_studentID2_studentID3.ipynb**. So, for example:
12345678_23456789_34567890.ipynb
- Answer all given questions: briefly describe what you have done and include your code's outputs as figures in the notebook.
- Try to understand the problem as much as you can. Provide evidence (experimental results, references to papers, etc.) to support your arguments.
- All illustrations (tables, plots, images, etc.) must be accompanied by analysis and explanation. All tables and plots must have variables' names and units, axes' names and legends. All results that are not understandable will be rejected.
- Only working code will be assessed. The TA's are not in a position to download libraries, delete code snippets or read documentation before reviewing your work.
- Please express your thoughts in a compact fashion. After all, it is not how many words you use that tells how well you understand the concepts.

Individual contribution. For the assignments, you will be assessed as groups. This means each student should contribute (more or less) equally to the submitted work. To make sure this happens, an individual contribution table is added to the end of each notebook. Filling out this table is mandatory. In principle, we will not use information from these tables to correct grades or otherwise change group assessment.

Programming. Python is the only programming language that is accepted for the assignments. Try to give meaningful names to your variables. The code should be readable and well-commented. It should be clear from the comments which function provides what functionality. You can only use python image processing libraries that are explicitly mentioned in the assignment description.

Late submissions are not allowed. Assignments that are submitted after the strict deadline will not be graded. In case of submission conflicts, TAs' system clock is taken as reference. We strongly recommend submitting well in advance, to avoid last minute system failure issues.

Plagiarism note. Keep in mind that plagiarism (submitted materials which are not your work, including using ChatGPT) is a serious crime and any misconduct shall be punished according to the university regulations.

6 Grading

Grades run from 1 to 10 (10 being the highest). The final grade is based on the following weighted parts:

- Exam 45%
- Lab assignments 50%
- Weekly lecture quizzes 5%

Each of the 5 quizzes counts for 1% (for a total of 5%).

The lab assignments part is subdivided as follows:

- The first four assignments contribute 50% to your lab grade. Assignments 1, 2, 3, 4 are weighted for 12%, 14%, 14%, 10% respectively.
- The rest 50% of your lab grade is based on the final project.

To pass the course, both the exam and the weighted average of the assignments should be scored at least a 5.5.

If you fail your Exam, you can take a resit. This resit grade replaces your Exam grade (45%). You cannot resit the lab assignments or the weekly quizzes.

Your final course grade will be rounded to the nearest 0.5 (except for 5.5, which does not exist under the UvA grading system).

7 Exam

All lecture slides and seminar material (exercises) will be material for the exam. The exam will be closed-book and will be done on-site, on paper, under regular UvA exam regulation.

Read the books and additional papers for background knowledge, but know that the slides and exercises are most important.

8 ChatGPT

The use of ChatGPT is **not** allowed. To achieve the intended learning outcome, hands on experience is very important. If we have solid evidence of you using ChatGPT, we will report it to the exam-committee.